PETRO QUANT

An Oil Market Intelligence
& Trading Strategy System

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What's Inside

MODULE 1	The 9 Market Signals — What moves oil prices, tracked automatically
MODULE 2	Market Mood Detection — Adapting to BULL, CHOPPY, and PANIC markets
MODULE 3	The Prediction Engine — AI that decides when to buy or sell
MODULE 4	News Intelligence — Reading 200+ articles daily so traders don't have to

Each module follows the same story arc

1 PROBLEM	→	2 SOLUTION	→	3 WHY BETTER	→	4 VS RIVALS
What pain does this solve? What are traders losing?		What does PetroQuant do? How does it work in plain English?		Why is our approach the smartest way to do it?		How do we compare to existing tools?

Four Modules

MODULE 1

The 9 Market Signals

- 9 real-time data sources
- Prices, USD, inventory, rigs…
- Automated daily refresh

Slides 3 – 6

MODULE 2

Market Mood Detection

- 3 market mood states
- BULL / CHOPPY / PANIC
- Auto risk adjustment

Slides 7 – 10

MODULE 3

The Prediction Engine

- AI buy/sell/hold signal
- Trained on history only
- 5-day price direction

Slides 11 – 14

MODULE 4

News Intelligence

- 200+ articles/day
- FinBERT AI scoring
- Supply/Demand/Geo topics

Slides 15 – 18

Traders Are Making Decisions With Incomplete Information

Oil doesn't move for one reason. It moves for nine — and most traders watch only one or two.

Most traders only track price.

But oil is actually moved by 9 interconnected forces — all happening simultaneously, every single day.

Missing even one of them means missed trades, wrong bets, and real money lost.

Tracking all 9 manually is impossible for a human. And most tools don't even try.

1 Watching Only Price

Charts show WHAT happened. They don't explain WHY — which means traders can't predict WHAT'S NEXT.

2 Ignoring the Dollar

Oil is priced in USD. When the dollar strengthens, oil demand globally drops. Missing this signal means trading against a headwind without knowing it.

3 Missing Inventory Data

The EIA inventory report is the most market-moving weekly release in oil. Most traders see the headline but never model the trend.

4 No Awareness of Speculator Bets

Hedge funds hold billions in oil futures. When their positioning reaches extremes, reversals follow. This signal is publicly available — and almost universally ignored.

PetroQuant Monitors All 9 Forces — Automatically, Every Day

Nine live data streams fused into one unified picture. Nothing slips through the cracks.

WTI Price

The US oil benchmark. The primary variable we predict. Every strategy starts here.

Brent Price

International benchmark. The gap between WTI and Brent reveals global vs US imbalances.

Oil Volatility (OVX)

The market's 'fear gauge'. High OVX = uncertainty. Low OVX = calm. Controls trade size.

US Dollar Index

Oil is priced in dollars. A rising dollar makes oil expensive globally → demand drops → price falls.

Crack Spread

Refinery profit margin. When refineries make good money, they buy more crude — driving prices up.

Speculator Bets (COT)

Hedge fund positioning. When everyone is piled into one side, a reversal is likely near.

Crude Oil Inventory

Weekly government data. Falling stockpiles = demand exceeds supply = bullish for price.

US Rig Count

Number of active drilling rigs. More rigs today = more supply in 3–6 months = future price pressure.

Petroleum Reserve (SPR)

Government reserve releases flood supply. Buying back supports it. Direct market intervention.

All 9 signals are checked every day. Missing or stale data is flagged. The system pauses and retries rather than making decisions on incomplete information.

Why Nine Signals Beat One — The Compounding Intelligence Effect

Each signal adds independently useful information. Together, they create a picture no single source can.

No Single Factor Is Enough

When oil dropped 40% in 2014, WTI price alone showed the fall — but the rig count, crack spread, AND dollar index had all been sending warning signals for weeks beforehand. Traders watching only price were caught off guard. PetroQuant would have seen all three.

Signals Confirm Each Other

When inventory is falling AND speculator bets are rising AND the dollar is weakening — that triple confirmation is far more reliable than any single indicator alone. PetroQuant looks for these convergences automatically.

Each Signal Has a Different Speed

Price moves daily. Inventories are reported weekly. Rig counts are monthly. PetroQuant aligns all three speeds onto one daily timeline — so nothing is lost and nothing is counted twice.

What This Means in Practice

- ✓ All 9 signals updated and aligned on the same daily calendar
- ✓ Data gaps are flagged — never silently filled with wrong values
- ✓ 24-hour caching means no wasted API calls, no rate limit failures
- ✓ Forward-fill logic ensures weekly/monthly data stays current
- ✓ Coverage report shows exactly which signals are working
- ✓ New signals can be added without changing any other code
- ✓ Historical data saved automatically for backtesting
- ✓ No Bloomberg required — built on open government APIs

How Our Data Coverage Compares to Existing Tools

The question isn't 'Do they have data?' — it's 'Do they have the RIGHT data, fused together?'

Tool	Oil Price	USD + Macro	Inventory Data	Speculator Bets
Bloomberg Terminal	Yes	Yes	Partial	Partial
RavenPack	Yes	Partial	NO	NO
Manual / Spreadsheet	Yes	Manual entry	Manual entry	Manual entry
Generic quant tools	Yes	Sometimes	Rarely	Rarely
PetroQuant	**YES**	**YES**	**YES**	**YES**

Key takeaway: Competitors either have pieces of the data or require manual work to combine it. PetroQuant is the only system that automatically fuses all 9 forces into one daily model.

The Market Flies in Three Different Modes — Most Systems Ignore This

A rule that works in calm weather is dangerous in a storm. Yet most trading strategies apply one rule to all conditions.

Imagine hiring a driver who drives the same speed in sun, fog, and blizzard.

That's what most trading systems do.

They apply the same buy/sell logic regardless of whether the market is in a calm uptrend, a directionless chop, or a full-blown panic.

The result: strategies that look great in backtests blow up the moment conditions change — which they always do.

No market stays in the same mood forever.

1 Bull Strategies Fail in Panics

A strategy calibrated for a calm, trending market will generate massive losses during a volatility spike. Oil crashed 300% in 2020. Most systems had no defense.

2 Panic Systems Miss Bull Runs

Being overly conservative in a strong bull market means leaving money on the table. The right aggression depends entirely on the current market environment.

3 Traders Adapt Too Slowly

Human traders adjust their style based on gut feel — with a lag of days or weeks. By the time they recognize a regime change, the opportunity (or the damage) is done.

4 No Tool Tells You What Mode You're In

Bloomberg shows price charts. Technical indicators show past patterns. None of them directly tell you: 'Right now, the market is in PANIC mode, reduce exposure.'

PetroQuant Reads the Market's Mood — and Adjusts Automatically

Three states. Automatic detection. Instant position adjustment. No human emotion involved.

BULL MODE

Low volatility. Prices trending upward.

What it looks like

Oil rises steadily. The news is broadly positive. Volatility is low. Speculators are building long positions.

What PetroQuant does

Trades at FULL size. Takes confident positions. This is the environment where the strategy performs best.

CHOPPY MODE

Moderate volatility. No clear direction.

What it looks like

Oil swings up and down without a clear trend. Mixed news — some bullish, some bearish. No strong conviction anywhere.

What PetroQuant does

Reduces position size to HALF. Still trades, but cautiously. Signal confidence is lower in choppy markets.

PANIC MODE

High volatility. Prices falling sharply.

What it looks like

Oil drops rapidly. Fear is very high. Headlines are driven by crises — COVID crash, Ukraine war, OPEC surprise.

What PetroQuant does

Cuts position size to ONE QUARTER. Capital preservation is the only priority. No strategy is reliable in this environment.

The Only System That Adapts Its Risk Automatically

Risk management that adjusts to market conditions — no human decision required in the heat of the moment.

How Position Sizing Works

BULL mood detected → **Trade at 100% of normal size**

CHOPPY mood detected → **Trade at 50% of normal size**

PANIC mood detected → **Trade at 25% of normal size**

The model also factors in signal confidence. A 70% probability prediction gets a bigger position than a 56% prediction — even if both say 'BUY'.

Final size = Direction × Mood Multiplier × Confidence

Why This Beats Manual Risk Management

No Emotional Bias

Human traders increase size when they're feeling confident — which often coincides with the end of a bull run. The system doesn't have emotions.

Instant Detection

The mood classification runs on every day of data. The day a PANIC starts, the system immediately adjusts. No three-meeting committee required.

Historically Proven

BULL, CHOPPY, and PANIC states are statistically distinct in terms of average returns and volatility — the model's classifications are mathematically grounded.

Protects Capital First

In PANIC mode, the goal shifts from profit to survival. A small loss in a crisis is far better than a large one — because the capital is still there to trade the recovery.

Does Your Tool Know What Mode the Market Is In Right Now?

If the answer is no — it's making decisions blind to market conditions.

Tool	Detects Market Mood?	Adjusts Trade Size?	Adapts in Real Time?	Protection in Panics?
Bloomberg / Terminals	**NO**	**NO — fixed rules**	**NO**	Manual only
Technical Indicators	**NO**	**NO**	**NO**	Stop-losses at best
Standard Quant Models	Rarely	Occasionally manual	Rarely	**Not systematic**
Human Trader	Partial	Emotional, delayed	Partial	Inconsistent
PetroQuant	**YES**	**YES — automatic**	**YES**	**YES — 75% size cut**

Key takeaway: No other tool systematically detects the market's mood AND automatically adjusts trading risk accordingly. This alone can be the difference between surviving a crash — and not.

Most Price Predictions Are Either Guesses or Rigged

The trading world is full of systems that 'backtested great' — then failed in real life. Here's why.

The central challenge of oil trading:

"Will oil be higher or lower next week?"

Sounds simple. It's not.

The wrong answer on a large position costs real money — fast.

Most systems answer this question using one of two broken approaches:

1. Human gut feel — emotional, inconsistent, doesn't scale.

2. Backtests that cheat — they train the model using future data, then claim it 'works'. In real trading, it fails immediately.

1 The 'Hindsight' Backtest Problem

If you train an AI model on all historical data — including the future — it looks brilliant. But it's already seen the answers. In live trading, it fails.

2 Technical Indicators Don't Predict

RSI, MACD, Bollinger Bands — these describe what happened. They don't integrate fundamental data: inventories, dollar, positioning, crack spreads.

3 One-Size-Fits-All Models

A model trained in 2018 market conditions makes poor decisions in 2024. Markets evolve. A static model becomes obsolete within months.

4 No Probability — Just 'Buy' or 'Sell'

Most signals are binary. But '51% probability of rising' and '80% probability' are very different situations. Trade size should reflect confidence, not just direction.

An AI That Learns From the Past and Only Predicts the Future

Walk-forward validation: the only honest way to test a trading strategy.

How it works in plain English:

1. We give the AI the first 2 years of oil history.
2. It learns patterns from those 2 years only.
3. We ask it to predict the next 3 months — data it has NEVER seen.
4. We measure how accurate those unseen predictions were.
5. We add those 3 months to the training set and repeat.
6. This continues until all history is covered.

The model is NEVER tested on data it trained on.
This is the gold standard in quantitative finance.

What the System Tells You

BUY Signal (>55% probability)

The AI is confident oil will be higher in 5 days. Position size scales with how confident — 70% gets a bigger trade than 56%.

SELL Signal (<45% probability)

The AI is confident oil will be lower in 5 days. Short position opened, again scaled by confidence level.

HOLD / Flat (45–55% probability)

The AI is not confident enough to act. No trade is placed. This 'no-trade zone' protects against marginal, low-edge situations.

The model also gives a 7-horizon forecast: 1 day, 7 days, 15 days … all the way to 180 days.

Predictions You Can Actually Trust — Because They Were Never Shown the Answers

Walk-forward validation makes the performance numbers real. Not theoretical. Not convenient.

Honest Performance Numbers

Because the model is only tested on data it never saw, the performance figures reflect how it would actually perform in live trading — not how well it can memorise history.

Adapts Every Quarter

The model is retrained with fresh data every 3 months. The relationship between the dollar and oil in 2022 is not identical to 2025. The model stays current.

Learns From Multiple Factors Together

Unlike simple indicators, the AI fuses all 9 signals simultaneously. It can discover relationships like: 'When inventory falls AND OVX is low AND USD weakens — that triple combination is 5x more predictive than any single factor alone.'

Confidence Scores Drive Position Size

The system doesn't just say BUY or SELL. It says '72% confident in a rise.' A higher confidence means a larger trade. This alone improves risk-adjusted returns significantly.

How Our Prediction Approach Compares

The key question: was the model ever tested on unseen data?

Prediction Approach	Uses Fundamental Data?	Honest Backtest?	Adapts Over Time?	Confidence Score?
Technical Analysis (TA)	**NO — price/volume only**	**Misleading**	**NO**	**NO**
Simple Quant / Linear Model	Partial	Often look-ahead bias	Rarely	**NO**
BlackBox AI vendor	Partial	Unknown — opaque	Unknown	Sometimes
Human Discretionary Trading	Partial / emotional	**Not applicable**	Slow/emotional	**NO**
PetroQuant	**YES — all 9 signals**	**YES — walk-forward**	**YES — quarterly**	**YES — probabilities**

Key takeaway: Most prediction systems either ignore fundamental data or use backtests that cheat. PetroQuant uses all 9 signals AND tests predictions only on data the model never trained on.

PROBLEM

200 Oil News Articles Are Published Every Day. Traders Read Five.

News drives oil prices. But the volume of news has made it impossible for any human to keep up.

The news you're missing is costing you money.

An OPEC statement can move oil 5% in an hour.
An EIA inventory surprise can shift $3/barrel instantly.
An Iranian sanctions headline hits before any chart does.

Reading 200 articles daily is not humanly possible.

And even if it were — a human reader can't consistently translate 'Iraqi production capacity constrained by infrastructure investment lag' into a precise sentiment score.

Generic AI tools make this worse — they don't understand oil.

1 Volume Is Impossible

Reuters, Bloomberg, EIA, OPEC, OilPrice.com, MarketWatch — all publishing simultaneously. Traders skim. Critical signals get missed.

2 Generic AI Gets It Wrong

'Oil inventories drew down 5 million barrels' — a generic sentiment tool sees the word 'down' and scores it negatively. It's actually bullish for prices.

3 News Moves Before Charts

Price action reflects news with a lag. A trader who reads the EIA report before the price moves has an edge. Most don't get there in time.

4 No Structure to the Noise

Not all oil news is equal. An OPEC supply decision matters more than a blog opinion. A morning article is more relevant than last week's. No tool weights these differences systematically.

PetroQuant Reads Every Article — and Understands Oil-Market Language

FinBERT: the AI model trained on financial text. One signal. Five topic breakdowns. Every morning.

What happens every morning at 6 AM:

Collect	200+ articles pulled from 4+ oil-specific news sources
Deduplicate	Near-identical articles removed — same story doesn't count twice
Score	FinBERT AI scores each article BULLISH, BEARISH, or NEUTRAL
Weight	Reuters article scores more than a random blog. Morning news scores more than yesterday's.
Classify	Each article tagged: Supply / Demand / Geopolitics / Macro / Price
Aggregate	All weighted scores combined into one master daily signal
Output	BULLISH, BEARISH, or NEUTRAL — with topic breakdown and divergence flag

What You Get — Sample Daily Output

Today's Signal:

BULLISH

Topic Breakdown:

Supply	-0.62	Oversupply fears dominant in supply news
Demand	+0.36	Positive demand signals from Asia
Geopolitics	-0.10	Minor tension in Middle East
Macro	+0.59	Positive economic backdrop
Price	+0.09	Neutral price commentary

Oil-Specific AI — Built for Energy Language, Not General Headlines

A tool that understands 'inventory drawdown = bullish' is fundamentally different from one that doesn't.

FinBERT Understands Financial Context

'Crude inventories drew down 5.3 million barrels' — FinBERT scores this correctly as BULLISH for oil prices. A generic sentiment tool flags the word 'down' as negative. This difference compounds across hundreds of articles every single day.

Source Authority Weighting

An EIA report or OPEC statement is weighted 3.5x higher than a generic news aggregator. A Reuters article counts more than an anonymous blog. Most tools give equal weight to everything — which dilutes the signal with noise.

Time Decay — Recency Matters

This morning's article has a 6-hour half-life. Yesterday's article has already lost most of its weight. Oil markets are forward-looking. Fresh intelligence matters more. No competitor applies this systematically.

Divergence Detection — The Early Warning

When Reuters says BEARISH but EIA data says BULLISH — PetroQuant flags this as 'CONTESTED'. High divergence historically precedes volatility spikes. This early warning signal is available at no extra cost — and no competitor provides it.

Five Topic Sub-Signals, Not One Flat Score

Knowing 'oil sentiment is neutral' is nearly useless. Knowing 'supply sentiment is -0.62 but demand is +0.59' tells a specific story: the market is worried about supply but expects demand to rescue it.

Deduplication — Quality Over Quantity

The same OPEC story appears across dozens of publications. PetroQuant removes near-duplicates before scoring. One event counts once — not thirty times.

How Our News Intelligence Compares to Every Alternative

The test: does the tool actually understand what oil market news MEANS?

Tool / Approach	Oil-Specific AI?	Source Weighting?	Topic Breakdown?	Divergence Alert?	Price?
RavenPack	**NO — generic**	**Yes (limited)**	Partial	**NO**	$50K–$500K/yr
MarketPsych / LSEG	**NO — generic**	**Yes**	Limited	**NO**	Enterprise only
Generic VADER/BERT	**NO — misreads oil**	**NO**	**NO**	**NO**	Free but wrong
Manual reading	Partial	Gut feel only	Manual	**NO**	$200K staff cost
PetroQuant	**YES — FinBERT**	**YES — 3-factor**	**YES — 5 topics**	**YES**	TBD

Key takeaway: No competitor at any price point provides oil-specific AI scoring + source weighting + time decay + topic breakdown + divergence detection — all in one daily signal.

PETROQUANT

Four modules. One unified oil intelligence system.

MODULE 1

9 Market Signals

Tracks all 9 forces that move oil — automatically, every day.

OUR EDGE

No competitor fuses all 9 into one model.

MODULE 2

Market Mood Detection

Identifies BULL, CHOPPY, or PANIC — adjusts trade size instantly.

OUR EDGE

No other tool adapts risk automatically to market conditions.

MODULE 3

The Prediction Engine

AI that learns only from the past and predicts the unseen future.

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Walk-forward validation — the gold standard. No look-ahead.

MODULE 4

News Intelligence

Reads 200+ oil articles daily. Scores, weights, and classifies them.

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Oil-specific FinBERT + divergence detection. Nothing like it exists.